

FOLD Progress

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April 4, 2025

New Dataset

- [bible-corpus](#) / [Paper](#)
- multilingual parallel corpus created from translations of the Bible
- 100 languages total, 34 overlap with FSI

New Baseline

- [lexical similarity](#)
- 69 Languages, 9 pair overlap with XNLI/56 pair overlap with bible-corpus
- computing normalized levenshtein distance from [AJSP](#) (10k basic vocabulary word lists) might be more rigorous

New Metric

Coherence

- No results yet, performance is an issue

Likelihood, `compute_overlaps` (600, 15, 15)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.429	0.126	-0.635	0.015	14
lex_sim	-0.974	0.000	-1.000	0.000	9
mut_int	-0.547	0.453	0.000	1.000	4

- All results are reported using Welch's method for estimating PSD

Likelihood, **k1_divergence_matrix**

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.384	0.175	0.743	0.002	14
lex_sim	0.986	0.000	0.976	0.000	9
mut_int	0.698	0.302	0.400	0.600	4

Likelihood, `mae_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.429	0.126	0.635	0.015	14
lex_sim	0.974	0.000	1.000	0.000	9
mut_int	0.547	0.453	0.000	1.000	4

Embeddings, `compute_overlaps`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.069	0.814	-0.362	0.204	14
lex_sim	-0.949	0.000	-0.976	0.000	9
mut_int	-0.547	0.453	0.000	1.000	4

Embeddings, `k1_divergence_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.354	0.214	0.631	0.016	14
lex_sim	0.952	0.000	0.839	0.005	9
mut_int	0.584	0.416	0.400	0.600	4

Embeddings, `mae_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.312	0.278	-0.606	0.022	14
lex_sim	-0.950	0.000	-0.976	0.000	9
mut_int	-0.547	0.453	0.000	1.000	4

Likelihood, `compute_overlaps` (600, 35, 35)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.143	0.419	-0.607	0.000	34
lex_sim	-0.801	0.000	-0.861	0.000	56
mut_int	-0.017	0.902	0.049	0.728	52

Likelihood, `k1_divergence_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.111	0.532	0.677	0.000	34
lex_sim	0.800	0.000	0.830	0.000	56
mut_int	-0.010	0.944	-0.114	0.422	52

Likelihood, `mae_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.143	0.419	0.607	0.000	34
lex_sim	0.801	0.000	0.861	0.000	56
mut_int	0.017	0.902	-0.049	0.728	52

Embeddings, `compute_overlaps`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.078	0.663	-0.539	0.001	34
lex_sim	-0.799	0.000	-0.921	0.000	56
mut_int	0.491	0.000	0.515	0.000	52

Embeddings, `k1_divergence_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.176	0.319	0.510	0.002	34
lex_sim	0.801	0.000	0.899	0.000	56
mut_int	-0.524	0.000	-0.533	0.000	52

Embeddings, `mae_matrix`

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.158	0.371	-0.500	0.003	34
lex_sim	-0.801	0.000	-0.917	0.000	56
mut_int	0.512	0.000	0.501	0.000	52

Discussion

Mutual Intelligibility

- With more languages from bible-corpus, mutual intelligibility is now significantly correlated for embeddings
- note: mutual intelligence has the opposite sign as lexical similarity and fsi, which is as expected

Lexical Similarity

- Very strong, *linear* correlation with fold
- present in all settings (all 3 metrics, likelihood and embedding, both datasets)

Next Steps

WALS